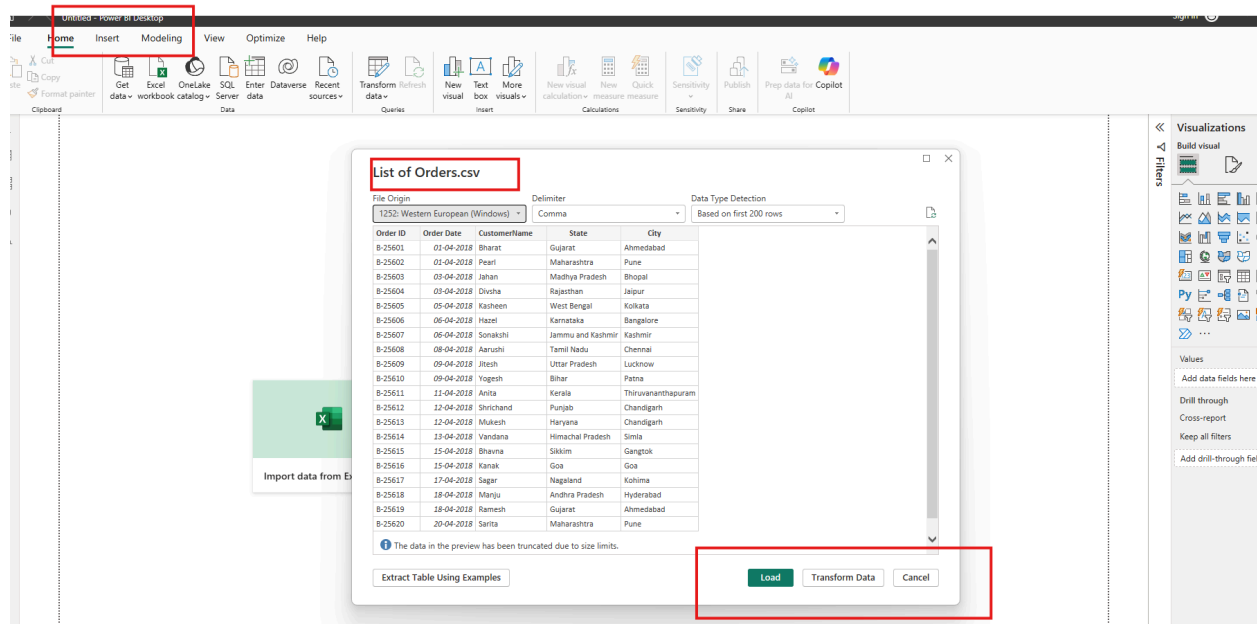
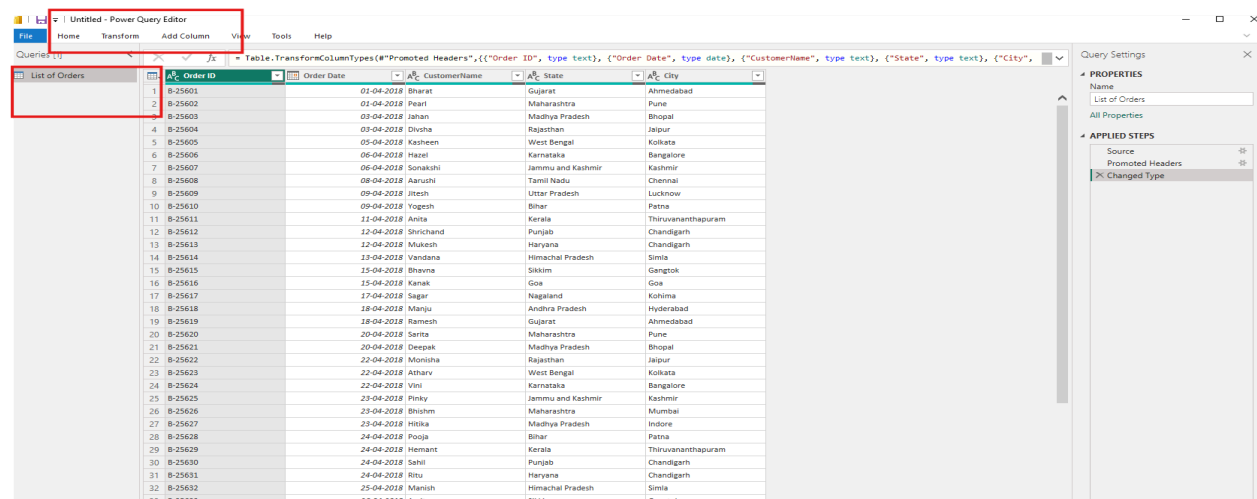


Data Import & Transformation

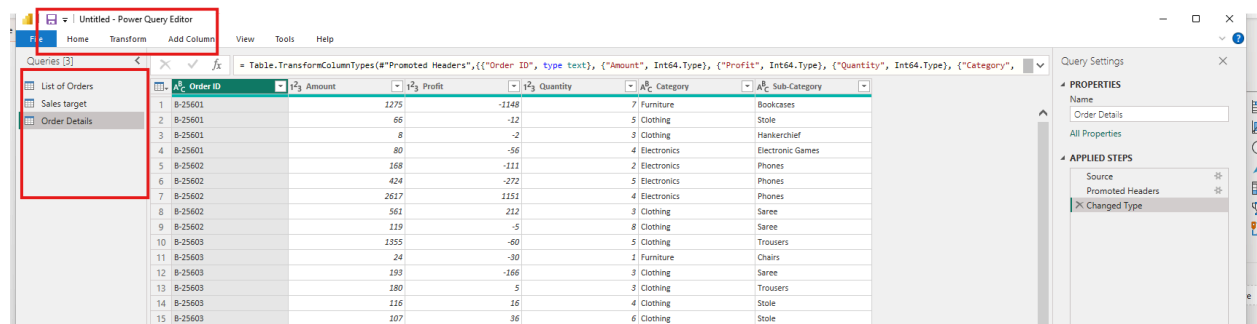
1. Data Import: Import List of Orders.csv into Power BI.



Open the list of orders in the Power Query Editor using the Transform Data option.



Import Order Details.csv and Sales Target.csv into Power Query Editor.



2. Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type default

Kept First Rows removed dupl...

Capitalized Each Word Custo...

Merged Columns Merged city...

Set the Order Date column to the Date data type.

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type default

Kept First Rows removed dupl...

Capitalized Each Word Custo...

Change the Amount and Target columns to fixed decimal numbers.

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type default

Kept First Rows removed dupl...

Capitalized Each Word Custo...

Data Cleaning and Transformation Performed on the Orders Table

1. Applied **Ascending Sort** to organize the Orders table.
2. Removed **60 blank and duplicate rows**, reducing the dataset to **500 rows**.
3. Changed the **Order Date** column data type to **Date**.
4. Changed the **Target** column data type to **Fixed Decimal Number**.
5. Changed the **Amount** column data type to **Fixed Decimal Number**

3. Text Formatting:

- Format the CustomerName column to proper case to ensure consistent capitalization.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh

4. Column Creation:

- Merge the City and State columns to create a new column named Location in the format: City, State

Order ID	Order Date	CustomerName	Location
B-25601	01-04-2018	Bharat	Ahmedabad, Gujarat
B-25602	01-04-2018	Pearl	Pune, Maharashtra
B-25603	03-04-2018	Jahan	Bhopal, Madhya Pradesh
B-25604	03-04-2018	Divsha	Jaipur, Rajasthan
B-25605	05-04-2018	Kasheen	Kolkata, West Bengal
B-25606	06-04-2018	Hazel	Bangalore, Karnataka
B-25607	06-04-2018	Sonakshi	Kashmir, Jammu and Kashmir
B-25608	08-04-2018	Aarushi	Chennai, Tamil Nadu
B-25609	09-04-2018	Jitesh	Lucknow, Uttar Pradesh
B-25610	09-04-2018	Yogesh	Patna, Bihar
B-25611	11-04-2018	Anita	Thiruvananthapuram, Kerala
B-25612	12-04-2018	Shrichand	Chandigarh, Punjab
B-25613	12-04-2018	Mukesh	Chandigarh, Haryana

Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	% Profit Margin
B-25601	1,275.00	127.50	7	Furniture	Bookcases	-90.04%
B-25602	168.00	-111.00	2	Electronics	Phones	-66.07%
B-25603	1,355.00	-60.00	5	Clothing	Trousers	-4.43%
B-25604	65.00	17.00	2	Clothing	T-shirt	26.15%
B-25605	75.00	0.00	7	Clothing	Saree	0.00%
B-25606	87.00	4.00	2	Clothing	Shirt	4.60%
B-25607	50.00	15.00	4	Clothing	Leggings	30.00%
B-25608	1,364.00	-1864.00	5	Furniture	Tables	-136.66%
B-25609	485.00	29.00	4	Electronics	Electronic Games	5.98%
B-25610	1,076.00	-38.00	4	Electronics	Printers	-3.53%
B-25611	160.00	-59.00	2	Clothing	Saree	-36.88%
B-25612	259.00	-55.00	2	Furniture	Chairs	-21.24%
B-25613	1,603.00	0.00	9	Clothing	Saree	0.00%

Observation

- Converted the **Customer Name** column to **Proper Case**, ensuring the first letter of each word is Capitalized
- Merged the **City** and **State** columns to create a new **Location** column, making location information easier to read and analyze.
- Created a **Profit Margin** custom column using the formula **Profit / Amount**, which shows the profit earned as a percentage of sales.
- The **Location** column simplifies regional analysis by combining city and state into a single field.
- The **Profit Margin** column helps identify which orders are more profitable and supports better business decision-making

5. Conditional Column

- Add a conditional column named Profit Status based on:
 - Profit < 0 → Loss
 - Profit = 0 → Break-Even
 - Profit > 0 → Profit

The screenshot shows a software interface with a data table and a dialog box for adding a conditional column.

Table Data (Top):

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
B-25602	168.00	-111	2	Electronics	Phones	-66.07%
B-25603	1,355.00	-60	5	Clothing	Trousers	-4.43%
B-25604	65.00	0	2	Clothing	T-shirt	26.15%
B-25605	75.00	0	7	Clothing	Saree	0.00%
B-25606	87.00	4	2	Clothing	Shirt	4.60%
B-25607	50.00	15	4	Clothing	Leggings	30.00%
B-25608	1,364.00	-1864	5	Furniture	Tables	-136.66%
B-25609	485.00	29	4	Electronics	Electronic Games	5.98%
B-25610	1,076.00	-38	4	Electronics	Printers	-3.53%
B-25611	160.00	-59	2	Clothing	Saree	-36.88%
B-25612	259.00	-55	2	Furniture	Chairs	-21.24%
B-25613	1,603.00	0	9	Clothing	Saree	0.00%
B-25614	494.00	54	4	Furniture	Bookcases	10.93%
B-25615	68.00	20	5	Clothing	Hankerchief	29.41%
B-25616	42.00	12	5	Clothing	Hankerchief	28.57%
B-25617	305.00	-270	5	Electronics	Electronic Games	-88.52%
B-25618	362.00	127	1	Furniture	Bookcases	35.08%
B-25619	353.00	90	8	Clothing	Saree	25.50%
B-25620	193.00	46	1	Electronics	Printers	23.83%

Add Conditional Column Dialog:

- New column name:** Profit Status
- Column Name:** Profit
- Operator:** is less than
- Value:** 0
- Output:** Loss
- Else:** Profit
- Operator:** equals
- Value:** 0
- Output:** Break-Even

Table Data (Bottom):

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss
B-25603	1,355.00	-60	5	Clothing	Trousers	-4.43%	Loss
B-25604	65.00	0	2	Clothing	T-shirt	26.15%	Break-Even
B-25605	75.00	0	7	Clothing	Saree	0.00%	Break-Even
B-25606	87.00	4	2	Clothing	Shirt	4.60%	Profit
B-25607	50.00	15	4	Clothing	Leggings	30.00%	Profit
B-25608	1,364.00	-1864	5	Furniture	Tables	-136.66%	Loss
B-25609	485.00	29	4	Electronics	Electronic Games	5.98%	Profit
B-25610	1,076.00	-38	4	Electronics	Printers	-3.53%	Loss
B-25611	160.00	-59	2	Clothing	Saree	-36.88%	Loss
B-25612	259.00	-55	2	Furniture	Chairs	-21.24%	Loss
B-25613	1,603.00	0	9	Clothing	Saree	0.00%	Break-Even
B-25614	494.00	54	4	Furniture	Bookcases	10.93%	Profit
B-25615	68.00	20	5	Clothing	Hankerchief	29.41%	Profit
B-25616	42.00	12	5	Clothing	Hankerchief	28.57%	Profit
B-25617	305.00	-270	5	Electronics	Electronic Games	-88.52%	Loss
B-25618	362.00	127	1	Furniture	Bookcases	35.08%	Profit
B-25619	353.00	90	8	Clothing	Saree	25.50%	Profit
B-25620	193.00	46	1	Electronics	Printers	23.83%	Profit

Properties Panel:

- APPLIED STEPS:**
 - Added Custom compute Profit...
 - Added Conditional Column Profit Status

Observations

1. Created a new **Profit Status** column based on the Profit values.
2. Orders with **negative profit** are marked as **Loss**.
3. Orders with **zero profit** are marked as **Break-Even**.
4. Orders with **positive profit** are marked as **Profit**, making it easier to identify profitable and unprofitable orders.

6. Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID

Amount

0 then

% Profit

Merge

Select a table and matching columns to create a merged table.

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07%	Loss
B-25603	1,355.00	-60	5	Clothing	Trousers	-4.43%	Loss
B-25604	65.00	17	2	Clothing	T-shirt	26.15%	Profit
B-25605	75.00	0	7	Clothing	Saree	0.00%	Break-Even

List of Orders

Order ID	Order Date	CustomerName	Location
B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat
B-25602	01-04-2018	Pearl	Pune,Maharashtra
B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh
B-25604	03-04-2018	Divsha	Jaipur,Rajasthan
B-25605	05-04-2018	Kasheen	Kolkata,West Bengal

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table.

OK Cancel

1,560.00 421 3 Clothing Trousers

133.00 12 5 Clothing Stole

on on 2 Furniture Furniture

File Home Transform Add Column View Tools Help

Queries [3]

Table.NestedJoin(#"Added Conditional Column Profit status using given data", {"Order ID"}, #"List of Orders", {"Order ID"}, "List of Orders", JoinKind.LeftOuter)

1 1,275.00 -1148 7 Furniture Bookcases -90.04% Loss

2 168.00 -111 2 Electronics Phones -66.07% Loss

3 1,355.00 -60 5 Clothing Trousers -4.43% Loss

4 65.00 17 2 Clothing T-shirt 26.15% Profit

5 75.00 0 7 Clothing Saree 0.00% Break-Even

6 87.00 4 2 Clothing Shirt 4.60% Profit

7 50.00 15 4 Clothing Leggings 30.00% Profit

8 1,964.00 -1864 5 Furniture Tables -136.66% Loss

9 485.00 29 4 Electronics Electronic Games 5.96% Profit

10 1,076.00 -38 4 Electronics Printers -3.53% Loss

11 160.00 -59 2 Clothing Saree -36.88% Loss

12 259.00 -55 2 Furniture Chairs -21.24% Loss

13 1,601.00 0 9 Clothing Saree 0.00% Break-Even

14 494.00 54 4 Furniture Bookcases 10.93% Profit

15 68.00 20 5 Clothing Hankerchief 29.41% Profit

16 42.00 12 5 Clothing Hankerchief 28.57% Profit

17 305.00 -270 5 Electronics Electronic Games -88.52% Loss

18 362.00 127 1 Furniture Bookcases 35.08% Profit

19 353.00 90 8 Clothing Saree 25.50% Profit

20 193.00 46 1 Electronics Printers 23.83% Profit

21 233.00 -10 5 Electronics Electronic Games -4.29% Loss

22 534.00 0 3 Clothing Saree 0.00% Break-Even

23 53.00 1 4 Clothing Stole 1.89% Profit

24 26.00 12 3 Clothing Hankerchief 46.15% Profit

25 97.00 29 2 Clothing Hankerchief 29.90% Profit

26 46.00 -14 1 Electronics Phones -30.43% Loss

27 55.00 -39 4 Clothing Stole -70.91% Loss

28 45.00 13 4 Clothing Skirt 28.89% Profit

29 1,560.00 421 3 Clothing Trousers 26.99% Profit

30

Order ID Order Date CustomerName Location

B-25601 01-04-2018 Bharat Ahmedabad,Gujarat

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted Headers

Changed Type default

Removed Duplicates order Id ...

Changed Type Amount to fix...

Added Custom compute Profi...

Changed Type1 hanged into ...

Added Conditional Column Pr...

Merged Queries

Name the merged output table as Orders Data.

	Sub-Category	% Profit Margin	Profit Status	List of Orders.Order Date	List of Orders.CustomerName	List of Orders.Location
1	Bookcases	-90.04%	Loss	01-04-2018	Bharet	Ahmedabad,Gujarat
2	Phones	-66.07%	Loss	01-04-2018	Pearl	Pune,Maharashtra
3	Trousers	-4.43%	Loss	03-04-2018	Jehan	Bhopal,Madhya Pradesh
4	T-shirt	26.15%	Profit	03-04-2018	Divsha	Jalpur,Rajasthan
5	Saree	0.00%	Break-Even	05-04-2018	Kasheen	Kolkata,West Bengal
6	Shirt	4.60%	Profit	06-04-2018	Hazel	Bangalore,Karnataka
7	Leggings	30.00%	Profit	06-04-2018	Sonakshi	Kashmir,Jammu and Kashmir
8	Tables	-136.66%	Loss	08-04-2018	Aarushi	Chennai,Tamil Nadu
9	Electronic Games	5.98%	Profit	09-04-2018	Itesh	Lucknow,Uttar Pradesh
10	Printers	-5.53%	Loss	09-04-2018	Yogesh	Patna,Bihar
11	Saree	-36.88%	Loss	11-04-2018	Anita	Thiruvananthapuram,Kerala
12	Chairs	-21.24%	Loss	12-04-2018	Shrichand	Chandigarh,Punjab
13	Saree	0.00%	Break-Even	12-04-2018	Mukesh	Chandigarh,Haryana
14	Bookcases	10.93%	Profit	13-04-2018	Vandana	Simla,Himachal Pradesh
15	Hankerschief	29.41%	Profit	15-04-2018	Bharna	Gangtok,Sikkim
16	Hankerschief	28.57%	Profit	15-04-2018	Kanak	Goa,Goa
17	Electronic Games	-88.52%	Loss	17-04-2018	Sagar	Kohima,Nagaland
18	Bookcases	35.08%	Profit	18-04-2018	Manju	Hyderabad,Andhra Pradesh
19	Saree	25.50%	Profit	18-04-2018	Ramesh	Ahmedabad,Gujarat
20	Printers	23.83%	Profit	20-04-2018	Sarita	Pune,Maharashtra
21	Electronic Games	-4.29%	Loss	20-04-2018	Deepak	Bhopal,Madhya Pradesh

Assignment

1. Select Merged the **List of Orders** table with the **Order Details** table using the **Order ID** column.
2. Set the **Left Outer Join** to keep all records from the **List of Orders** table and match the related records from the **Order Details** table.
3. Named the merged query **Orders Data**
4. Expanded the merged table to extract the required columns from **Order Details** and created a separate output table named **Orders Data**.

Observations

1. The **Left Outer Join** kept all records from the **List of Orders** table, even if no matching **Order Details** existed.
2. Matching columns from the **Order Details** table were successfully expanded into the merged table.
3. The final **Orders Data** table combines order information and order details in one place.
4. The merged dataset is ready for reporting, analysis, and data modeling in Power BI.

7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
 - Replacing missing values
 - Removing records
 - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

Handling Missing Data & Duplicate Data

Assignment:

1. Checked all tables and columns for **missing (null) values** and **duplicate records**.
2. Found **60 null values** in the **Order ID** column and removed those 60 records because an Order ID is mandatory for identifying each order.
3. Verified the remaining data for duplicate rows based on business logic.
4. No other missing values or duplicate records were found in the dataset.

Observations

1. Identified and removed **60 records with null Order IDs** to ensure data quality.
2. **Order ID** is a key field, so records with missing Order IDs were removed.
3. All tables and columns were checked for additional missing values and duplicates.
4. No other missing data or duplicate records were found, leaving the dataset clean and ready for analysis.

8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending.

Order Data

List of Orders

Order Details

Category	Sub-Category	% Profit Margin	Profit Status	List of Orders.Order Date	List of Orders.CustomerName	List of Orders.Location
1	2 Furniture	Chairs	27.78% Profit	31-03-2019	Hitika	Indore,Madhya Pradesh
2	1 Clothing	Skirt	33.33% Profit	30-03-2019	Bhishm	Mumbai,Maharashtra
3	3 Electronics	Accessories	9.76% Profit	29-03-2019	Pinky	Kashmir,Jammu and Kashmir
4	2 Clothing	Hankercchief	12.37% Profit	28-03-2019	Vini	Bangalore,Karnataka
5	3 Clothing	Leggings	20.00% Profit	28-03-2019	Atharv	Kolkata,West Bengal
6	1 Clothing	Kurti	16.67% Profit	28-03-2019	Monisha	Jaipur,Rajasthan
7	6 Clothing	Stole	32.89% Profit	27-03-2019	Deepak	Bhopal,Madhya Pradesh
8	3 Clothing	Stole	43.67% Profit	27-03-2019	Manju	Hyderabad,Andhra Pradesh
9	2 Clothing	T-shirt	14.43% Profit	27-03-2019	Ramesh	Ahmedabad,Gujarat
10	1 Clothing	Saree	-3.03% Loss	27-03-2019	Sarita	Pune,Maharashtra
11	3 Clothing	Stole	27.50% Profit	27-03-2019	Sagar	Kohima,Nagaland
12	2 Clothing	Hankercchief	45.45% Profit	26-03-2019	Bhavna	Gangtok,Sikkim
13	2 Clothing	T-shirt	39.53% Profit	26-03-2019	Mukesh	Chandigarh,Haryana
14	7 Clothing	Saree	47.06% Profit	26-03-2019	Vandana	Simla,Himachal Pradesh
15	2 Clothing	Saree	25.58% Profit	26-03-2019	Shrichand	Chandigarh,Punjab
16	6 Clothing	Kurti	40.68% Profit	26-03-2019	Kanak	Goa,Goa
17	4 Electronics	Electronic Games	-30.14% Loss	25-03-2019	Anita	Thiruvananthapuram
18	3 Clothing	Leggings	18.60% Profit	24-03-2019	Yogesh	Patna,Bihar
19	2 Clothing	Stole	5.26% Profit	23-03-2019	Jitesh	Lucknow,Uttar Pradesh

Properties

Order Data

All Properties

Applied Steps

Source

Promoted Headers

Changed Type default

Removed Duplicates: order Id ...

Changed Type Amount to fixe...

Added Custom compute Profi...

Changed Type1 hanged into lo...

Added Conditional Column Pr...

Merged Queries list of order...

Expanded List of Orders creat...

Order date in Decending rde...

- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

The screenshot shows the same data table as above, but with filters applied. The 'List of Orders.Location' column is highlighted with a red box, indicating it is the primary filter. The 'List of Orders.Location' column is filtered to show only 'Chennai,Tamil Nadu'. The 'List of Orders.Order Date' column is also highlighted with a red box, indicating it is the secondary filter. The table is sorted by Order Date in descending order.

Category	Sub-Category	% Profit Margin	Profit Status	List of Orders.Order Date	List of Orders.CustomerName	List of Orders.Location
1	ure	Bookcases	-94.15% Loss	22-03-2019	Aarushi	Chennai,Tamil Nadu
2	ure	Furnishings	32.82% Profit	14-02-2019	Aarushi	Chennai,Tamil Nadu
3	ng	Stole	18.18% Profit	09-02-2019	Kalyani	Chennai,Tamil Nadu
4	ng	Kurti	21.43% Profit	15-11-2018	Akshay	Chennai,Tamil Nadu
5	ng	Stole	25.00% Profit	21-09-2018	Dinesh	Chennai,Tamil Nadu
6	ng	Saree	0.00% Break-Even	11-07-2018	Surabhi	Chennai,Tamil Nadu
7	ng	Kurti	-95.40% Loss	23-06-2018	Amisha	Chennai,Tamil Nadu
8	ure	Tables	-136.66% Loss	08-04-2018	Aarushi	Chennai,Tamil Nadu

: Sorting and Filtering Data

Steps.

- Sorted **Order Date** in **Descending** order.
- Filtered the **State** column.
- Selected **Tamil Nadu**.
- Displayed only Tamil Nadu orders.

Simple Observations

- Latest orders appear first after sorting.
- The filter shows only Tamil Nadu orders.
- Sorting and filtering simplify data analysis.
- The data is easier to review and analyze.

9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
 - Count of Order ID

Table: Order ID, Count of order Id

Order ID	Count of order Id
B-25601	1
B-25602	1
B-25603	1
B-25604	1
B-25605	1
B-25606	1
B-25607	1
B-25608	1
B-25609	1
B-25610	1
B-25611	1
B-25612	1
B-25613	1
B-25614	1
B-25615	1
B-25616	1
B-25617	1
B-25618	1
B-25619	1
B-25620	1
B-25621	1
B-25622	1
B-25623	1
B-25624	1
B-25625	1
B-25626	1
B-25627	1
B-25628	1
B-25629	1
B-25630	1
B-25631	1
B-25632	1
B-25633	1
B-25634	1
B-25635	1
B-25636	1

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type default
- Removed Duplicates order Id ...
- Changed Type Amount to fix...
- Added Custom compute Profit...
- Changed Type1 handed into ...
- Added Conditional Column Profit status using given data
- Added Conditional Column Profit status using given data

Average Profit by Category

Table: Category, Average Profit by Category

Category	Average Profit by Category
Furniture	20.44318182
Electronics	-19.37894797
Clothing	15.29022082

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type default
- Removed Duplicates order Id ...
- Changed Type Amount to fix...
- Added Custom compute Profit...
- Changed Type1 handed into ...
- Added Conditional Column Profit status using given data
- Added Conditional Column Profit status using given data

Total Amount by Sub-Category

Table: Sub-Category, Total Amount by Sub-Category

Sub-Category	Total Amount by Sub-Category
Bookcases	25800
Phones	11691
Trimmers	7209
T-shirt	3205
Saree	17831
Shirt	2535
Leggings	851
Tables	3967
Electronic Games	11942
Printers	16822
Chairs	5724
Hankierchief	4362
Stole	7583
Skirt	822
Furnishings	4295
Accessories	9472
Kurti	1154

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type default
- Removed Duplicates order Id ...
- Changed Type Amount to fix...
- Added Custom compute Profit...
- Changed Type1 handed into ...
- Added Conditional Column Profit status using given data
- Added Conditional Column Profit status using given data

- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Query Editor: Table Group("Changed Type default" ("Month of Order Date"), [{"Count", each List.Sum([Target]), type nullable number}]})

Month of Order Date	Count
01-04-2018	31400
01-05-2018	31500
01-06-2018	31800
01-07-2018	31800
01-08-2018	31900
01-09-2018	34000
01-10-2018	36100
01-11-2018	36300
01-12-2018	36400
01-01-2019	42500
01-02-2019	43600
01-03-2019	43800

Query Settings: Name: the total Target by Month of order date

APPLIED STEPS: Source, Promoted Headers default, Changed Type default, Target by Month of order date

Grouping and Aggregating Data

Steps Duplicated the **Order Details** table.

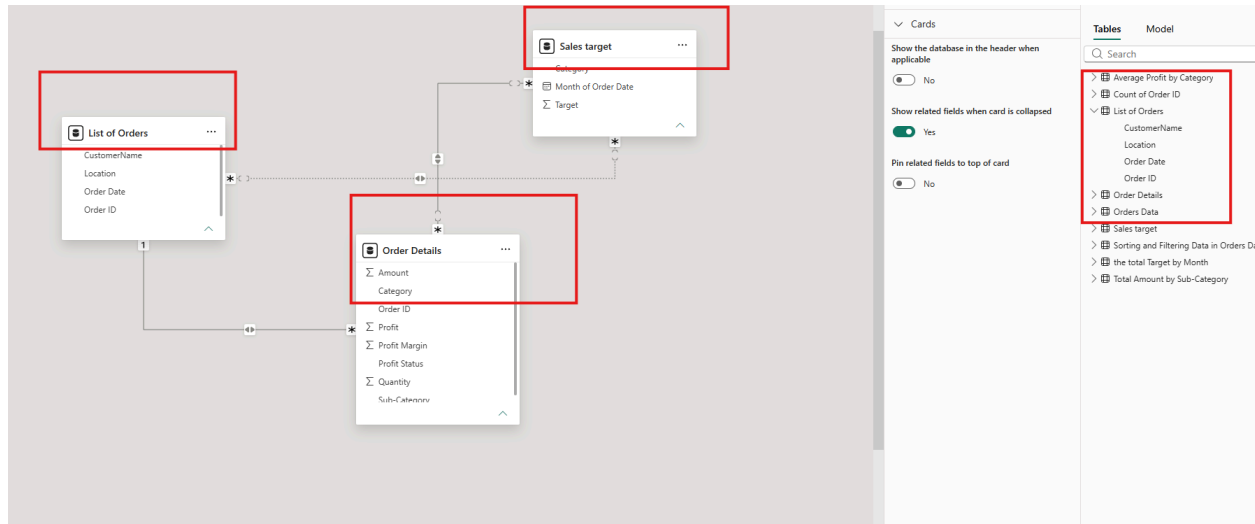
1. Grouped the data to find **Count of Order ID**, **Average Profit by Category**, and **Total Amount by Sub-Category**.
2. Duplicated the **Sales Target** table.
3. Grouped the data to calculate the **Total Target by Month (Order Date)**.

Simple Observations

1. Order count shows the total number of orders.
2. Average profit helps compare category performance.
3. Total amount shows sales for each sub-category.
4. Monthly target aggregation helps analyze target performance by month.

10. Data Modeling:

- Establish a relationship between:
 - List of Orders and Order Details using Order ID
 - Establish a relationship between:
 - Order Details and Sales Target using Category
2. Use Manage Relationships to ensure relationships are active and correctly defined.



Data Modeling

Steps Performed

1. Created a relationship between **List of Orders** and **Order Details** using **Order ID**.
2. Created a relationship between **Order Details** and **Sales Target** using **Category**.
3. Used **Manage Relationships** to verify that both relationships are **active**.
4. Confirmed the relationships were correctly defined for data analysis.

Simple Observations

1. **Order ID** links the List of Orders and Order Details tables.
2. **Category** links the Order Details and Sales Target tables.
3. Active relationships allow data to flow correctly between tables.
4. The data model is ready for reporting and analysis in Power BI.

Overall Observations

1. Successfully imported and transformed all three datasets in **Power Query**.
2. Cleaned the data by correcting data types, formatting text, creating new columns, and removing invalid records.
3. Merged tables, grouped data, and applied sorting and filtering to make the data easier to analyze.
4. Created relationships between tables, resulting in a well-structured data model ready for reporting and visualization in Power BI.

Summary

- Imported **List of Orders**, **Order Details**, and **Sales Target** into Power BI.
- Cleaned and transformed the data by setting correct data types, formatting customer names, creating **Location**, **Profit Margin**, and **Profit Status** columns, and removing **60 records with null Order IDs**.
- Merged tables, grouped and aggregated data, and applied sorting and filtering for better analysis.
- Built active relationships using **Order ID** and **Category**, making the dataset ready for creating dashboards and performing accurate business analysis.